METRIC
DOD-P-24631/3A(SH)
16 October 1987
SUPERSEDING
DOD-P-24631/3(SH)
13 April 1984

MILITARY SPECIFICATION SHEET

PAINT, EPOXY, WHITE, NAVY FORMULA 186, TYPE III (METRIC)

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the paint described herein shall consist of this specification and the latest issue of DOD-P-24631(SH).

Formula: This formula covers white epoxy paint designated Navy formula 186 for exterior use. The paint shall consist of the ingredients specified in the quantities specified.

AMSC N/A FSC 8010
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...ual amount of residue retained when the dry zinc pigment is subjected to the ...

TABLE I. Navy formula 186 - component A.

	Component A	Pounds	Kilograms	Gallons[1/]	Liters[1/]
Curing agent	Scotch Weld 2216A (clear amber), 3M Company	2118.4	960.9	259.03	980.4
Titanium dioxide	ASTM D 476, type III or IV	1124.9	510.3	33.7	127.6
Flatting agent	Silcron G601, Glidden Pigments	69.0	31.3	3.94	14.9
Aromatic solvent	HI SOL 10, Ashland	93.7	42.5	12.73	48.2
n-Butyl alcohol	TT-B-848	400.0	181.4	59.10	223.7
Ethylene glycol mono ethyl ether	TT-E-781	400.0	181.4	51.49	194.9
Ethylene glycol mono butyl ether	TT-E-776	133.8	60.7	17.78	67.3
Thickener	Bentone 38, NL Ind.	59.3	26.9	3.95	15.0
Thickener activator	Propylene carbonate, Jefferson Chemical	20.7	9.4	2.09	7.9
Dispersant	Tenlo 70, Diamond Shamrock	2.8	1.3	0.34	1.3
		4422.6	2006.1	444.15	1681.2

1/ Provided for guidance only - actual values are dependent on variation in raw materials.

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TABLE II. Navy formula 186 - component B.

	Component B	Pounds	Kilograms	Gallons 1/	Liters 1/
Epoxy resin	Scotch Weld 2216B Clear, 3M Company	1923.5	872.5	200.43	758.6
Titanium dioxide	ASTM D 476, type III or IV	1532.1	695.0	45.9	173.7
Flatting agent	Silcron G601, Glidden Pigments	191.6	86.9	10.93	41.4
Aromatic solvent	HI SOL 10, Ashland	127.6	57.9	17.47	66.1
n-Butyl acetate	TT-B-848	544.8	247.1	73.93	279.8
Ethylene glycol mono ethyl ether	TT-E-781	544.8	247.1	70.12	265.4
Ethylene glycol mono butyl ether	TT-E-776	182.2	82.6	24.21	91.6
Thickener	Bentone 38, NL Ind.	56.4	25.6	3.75	14.2
Thickener activator	Propylene Carbonate, Jefferson Chemical	18.8	8.5	1.89	7.2
Dispersant	Tenlo 70, Diamond Shamrock	3.8	1.7	0.48	1.7
		5125.6	2324.9	449.11	1699.7

1/ Provided for guidance only - actual values are dependent on variation in raw materials.

NOTES:

1. The curing agent shall be Scotch Weld 2216A (transparent) available from 3M Company, 3M Center, 2501 Hudson Road, St. Paul, MN, or equivalent (see figure 1 of this specification sheet).
2. The solvent shall be any 97-99 percent aromatic with a flash point of 100-110 degrees Fahrenheit (°F) (38-43 degrees Celsius (°C)).
3. The thickener shall be "Bentone 38" available from N.L. Industries, Inc., Hightstown, NJ, or equivalent.
4. The thickener activator shall be "propylene carbonate" available from Jefferson Chemical, Shaker Heights, OH, or equivalent.
5. The dispersant shall be "Tenlo 70" available from Diamond Shamrock, Morristown, NJ, or equivalent.
6. The flatting agent shall be "Silcron G601" available from Glidden-Durkee Division, S.C.M. Corporation, Baltimore, MD, or equivalent.
7. The epoxy resin shall be Scotch Weld 2216B (clear amber) available from 3M Company, 3M Center, 2501 Hudson Road, St. Paul, MN, or equivalent (see figure 2 of this specification sheet).

Use of alternative ingredients in this formula shall have prior approval by the Naval Sea Systems Command. Approval will be based on review of data showing equivalent physical and chemical characteristics to the specified ingredient. It will be necessary to demonstrate that the paint compound made using the alternative ingredient will conform to all requirements of this specification.

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TABLE III. Quantitative requirements.

Requirements	Component A		Component B		Mixed component	
	Min	Max	Min	Max	Min	Max
Pigment:						
Percent by mass of paint	23.4	27.4	27.9	31.9		
Percent by volume of paint	5.6	9.6	8.2	12.2		
Volatiles:						
Percent by mass of paint	21.7	25.7	25.7	29.7		
Percent by volume of paint	30.2	34.2	39.8	43.8		
Nonvolatiles:						
Percent by mass of paint	48.9	52.9	40.4	44.4		
Percent by volume of paint	58.2	62.2	46.0	50.0		
Consistency, Krebs units					60.0	——
Mass per unit volume:						
lbs/gallon	9.71	10.21	11.16	11.66	10.44	10.94
Kg/L	1.163	1.223	1.338	1.398	1.250	1.310
Fineness of grind (Hegman)					6.0	——
Flash point, Pensky-Martens						
°C					27.0	——
°F					80.0	——
Set-to-touch time:						
hours at 23°C (73°F)					6.0	8.0
Dry-hard hours at 23°C (73°F)					——	24.0
Titanium dioxide,						
percent of pigment	75.4	——	71.1	——		
Pot life, hours at 23°C (73°F)					6.0	——
Gloss, 16°C (60°F) specular,					——	10.0
percent						
Contrast ratio					0.9	

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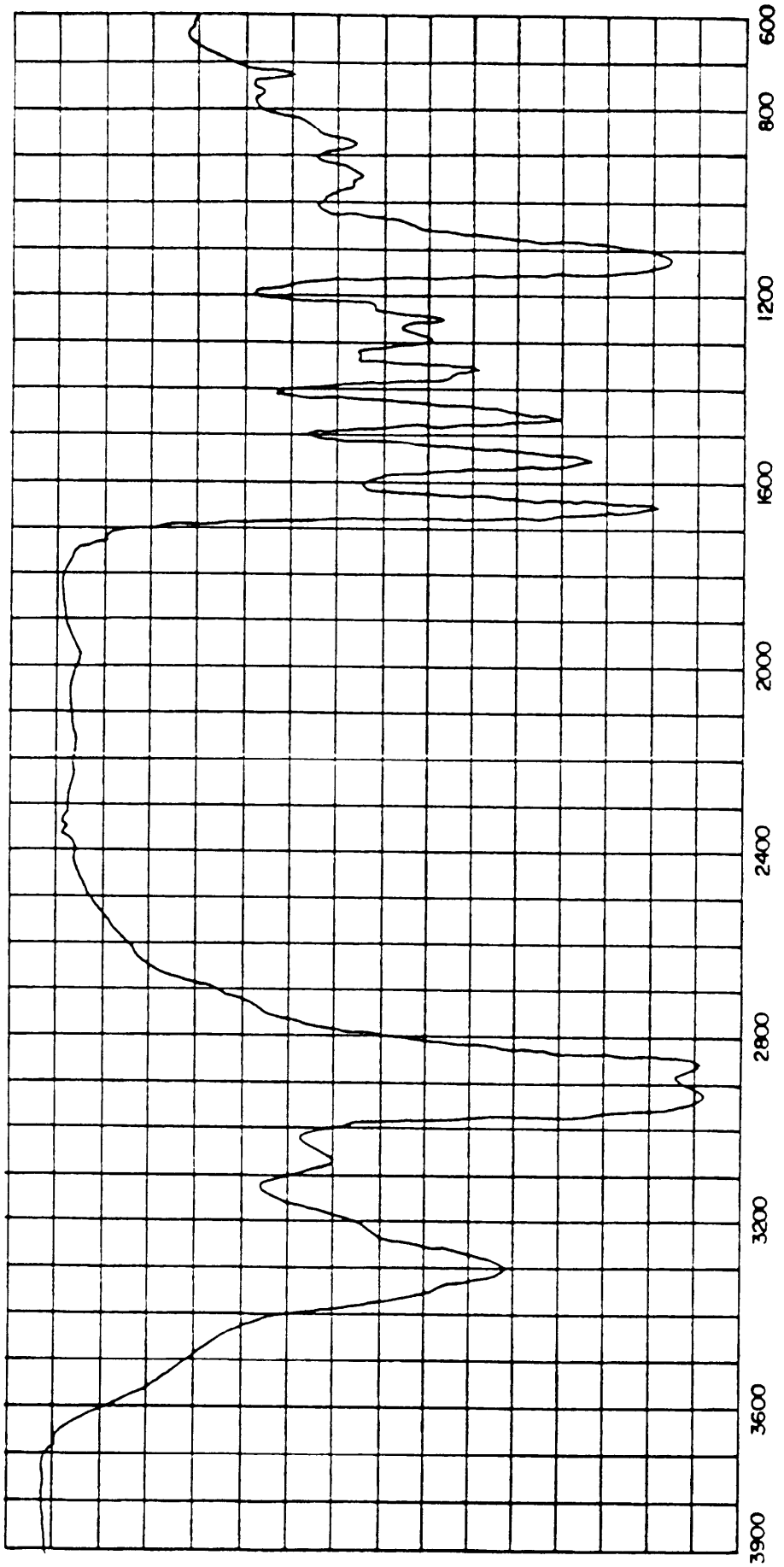


FIGURE 1. Infrared spectrogram of Scotch Weld 2216A (aliphatic amine).

SH 12143

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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